

**Aim:** To design and represent an Entity Relationship (ER) diagram for a real-world database system using the draw.io tool by identifying entities, attributes, keys, relationships and cardinalities.

### **1. Identify all entity types and entity sets.**

**Researcher-** Represents all researchers working in the university.

**Department-** Represents all departments in the university.

**Research Project-** Represents all research projects undertaken by the university.

**Student-** Represents all students working under researchers.

**Publication-** Represents all publications produced from research projects.

**Project Assistant-** Represents all assistants associated with research projects.

### **2. Identify strong and weak entity sets.**

Strong and Weak Entity Sets

- **Strong Entity Sets:**

- Researcher
- Department
- Research Project
- Student
- Publication

- **Weak Entity Set:**

- Project Assistant

**Project Assistant** is a weak entity because it is identified only within the context of a particular Research Project.

### **3. Identify the owner entity and partial key of the weak entity.**

**Weak Entity:** Project Assistant

**Owner Entity:** Research Project

**Partial Key:** Assistant\_Name

The complete identification of a Project Assistant is: **Project\_ID + Assistant\_Name**

#### 4. Simple attributes.

The simple attributes in the University Research Management System are:

- **Researcher:** Researcher\_ID, Name, Date\_of\_Birth, Email
- **Department:** Department\_ID, Department\_Name, Location
- **Research Project:** Project\_ID, Project\_Title, Start\_Date, End\_Date, Budget
- **Student:** Student\_ID, Name, Email, Date\_of\_Birth
- **Publication:** Publication\_ID, Title, Publication\_Date, DOI, Publisher
- **Project Assistant:** Assistant\_Name, Role, Joining\_Date

#### 5. Identify composite attributes.

The composite attribute in the system is:

##### 1. Researcher → Address

- Street
- City
- State
- PIN\_Code

**Address** is a composite attribute because it can be divided into smaller component attributes.

**Name** is treated as a simple attribute because the given problem statement does not specify First\_Name, Middle\_Name, and Last\_Name.

#### 6. Identify single-valued attributes.

##### Researcher

- Researcher\_ID
- Name
- Date\_of\_Birth
- Email
- Address

##### Department

- Department\_ID
- Department\_Name
- Location

## Research Project

- Project\_ID
- Project\_Title
- Start\_Date
- End\_Date
- Budget

## Student

- Student\_ID
- Name
- Email
- Date\_of\_Birth

## Publication

- Publication\_ID
- Title
- Publication\_Date
- DOI
- Publisher

## Project Assistant

- Assistant\_Name
- Role
- Joining\_Date

- **7. Identify multi-valued attributes.**

The multi-valued attributes in the system are:

- **1. Researcher → Phone\_Number**  
A researcher can have multiple phone numbers.
- **2. Research Project → Keyword**  
A research project can have multiple keywords.

## **8. Identify stored attributes.4**

### **Researcher**

- Researcher\_ID
- Name
- Date\_of\_Birth
- Email
- Phone\_Number
- Address
  - Street
  - City
  - State
  - PIN\_Code

### **Department**

- Department\_ID
- Department\_Name
- Location

### **Research Project**

- Project\_ID
- Project\_Title
- Start\_Date
- End\_Date
- Budget
- Keyword

### **Student**

- Student\_ID
- Name
- Email
- Date\_of\_Birth

### **Publication**

- Publication\_ID

- Title
- Publication\_Date
- DOI
- Publisher

### **Project Assistant**

- Assistant\_Name
- Role
- Joining\_Date

## **9. Identify derived attributes.**

### **Derived Attribute – Age**

**Age** is derived from **Date\_of\_Birth**.

Therefore, Age does not need to be stored separately in the database.

In the ER diagram, **Age** is represented using a **dashed oval**.

## **10. Identify key attributes.**

| <b>Entity</b>            | <b>Key Attribute</b>        | <b>Key Status</b> |
|--------------------------|-----------------------------|-------------------|
| <b>Researcher</b>        | Researcher_ID               | Primary Key       |
| <b>Department</b>        | Department_ID               | Primary Key       |
| <b>Research Project</b>  | Project_ID                  | Primary Key       |
| <b>Student</b>           | Student_ID                  | Primary Key       |
| <b>Publication</b>       | Publication_ID              | Primary Key       |
| <b>Project Assistant</b> | Assistant_Name              | Partial Key       |
| <b>Project Assistant</b> | Project_ID + Assistant_Name | Composite Key     |

### 11. Identify candidate keys and select appropriate primary keys.

| Entity            | Candidate Key               | Selected Primary Key        |
|-------------------|-----------------------------|-----------------------------|
| Researcher        | Researcher_ID               | Researcher_ID               |
| Department        | Department_ID               | Department_ID               |
| Research Project  | Project_ID                  | Project_ID                  |
| Student           | Student_ID                  | Student_ID                  |
| Publication       | Publication_ID              | Publication_ID              |
| Project Assistant | Project_ID + Assistant_Name | Project_ID + Assistant_Name |

### 12. Identify at least one composite key.

The composite key is: **Project Assistant** → **Project\_ID + Assistant\_Name**

**Assistant\_Name** alone cannot uniquely identify an assistant because the same assistant name may occur in different projects.

Therefore: **Composite Key = (Project\_ID, Assistant\_Name)**

Another composite key occurs in the relational table created for the M:N **WORKS\_ON** relationship: **(Researcher\_ID, Project\_ID)**

### 13. Identify appropriate foreign keys after converting the ER model into relations.

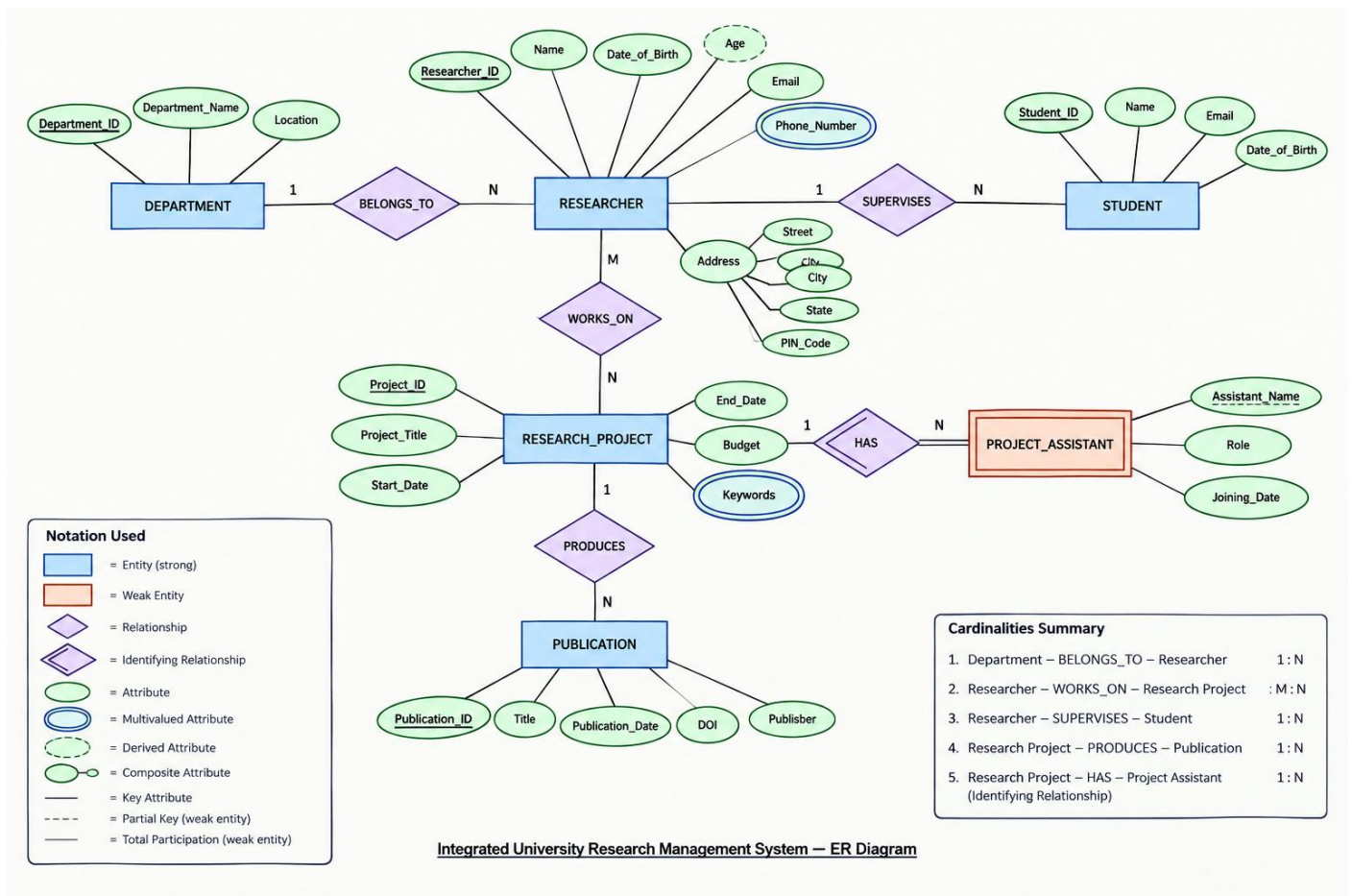
| Relation    | Foreign Key   | References                   |
|-------------|---------------|------------------------------|
| RESEARCHER  | Department_ID | DEPARTMENT(Department_ID)    |
| WORKS_ON    | Researcher_ID | RESEARCHER(Researcher_ID)    |
| WORKS_ON    | Project_ID    | RESEARCH_PROJECT(Project_ID) |
| STUDENT     | Researcher_ID | RESEARCHER(Researcher_ID)    |
| PUBLICATION | Project_ID    | RESEARCH_PROJECT(Project_ID) |

| Relation                 | Foreign Key   | References                   |
|--------------------------|---------------|------------------------------|
| <b>PROJECT_ASSISTANT</b> | Project_ID    | RESEARCH_PROJECT(Project_ID) |
| <b>RESEARCHER_PHONE</b>  | Researcher_ID | RESEARCHER(Researcher_ID)    |
| <b>PROJECT_KEYWORD</b>   | Project_ID    | RESEARCH_PROJECT(Project_ID) |

**14. Identify all major relationships between the entities.**

| Relationship         | Entities Involved                    | Cardinality | Meaning                                                                                                                                                      |
|----------------------|--------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BELONGS_TO</b>    | Department ↔ Researcher              | 1 : N       | One department has many researchers; each researcher belongs to one department.                                                                              |
| <b>WORKS_ON</b>      | Researcher ↔ Research Project        | M : N       | A researcher can work on many projects, and a project can involve many researchers.                                                                          |
| <b>SUPERVISES</b>    | Researcher ↔ Student                 | 1 : N       | One researcher can supervise many students; each student works under one researcher.                                                                         |
| <b>PRODUCES</b>      | Research Project ↔ Publication       | 1 : N       | One project can produce many publications; each publication belongs to one project.                                                                          |
| <b>HAS_ASSISTANT</b> | Research Project ↔ Project Assistant | 1 : N       | One project can have many assistants; each assistant belongs to one project. This is an identifying relationship because Project Assistant is a weak entity. |

## 15. Draw the complete ER diagram using standard ER notation.



## 16. Prepare a table showing each attribute and its attribute type and key status.

| Entity     | Attribute     | Attribute Type   | Value        | Key Status  |
|------------|---------------|------------------|--------------|-------------|
| Researcher | Researcher_ID | Simple           | Single       | Primary Key |
| Researcher | Name          | Simple           | Single       | Non-key     |
| Researcher | Date_of_Birth | Simple           | Single       | Stored      |
| Researcher | Age           | Derived          | Single       | Derived     |
| Researcher | Email         | Simple           | Single       | Non-key     |
| Researcher | Phone_Number  | Simple           | Multi-valued | Non-key     |
| Researcher | Address       | Composite        | Single       | Non-key     |
| Researcher | Street        | Simple component | Single       | Non-key     |

| Entity           | Attribute        | Attribute Type   | Value        | Key Status         |
|------------------|------------------|------------------|--------------|--------------------|
| Researcher       | City             | Simple component | Single       | Non-key            |
| Researcher       | State            | Simple component | Single       | Non-key            |
| Researcher       | PIN_Code         | Simple component | Single       | Non-key            |
| Department       | Department_ID    | Simple           | Single       | <b>Primary Key</b> |
| Department       | Department_Name  | Simple           | Single       | Non-key            |
| Department       | Location         | Simple           | Single       | Non-key            |
| Research Project | Project_ID       | Simple           | Single       | <b>Primary Key</b> |
| Research Project | Project_Title    | Simple           | Single       | Non-key            |
| Research Project | Start_Date       | Simple           | Single       | Stored             |
| Research Project | End_Date         | Simple           | Single       | Stored             |
| Research Project | Budget           | Simple           | Single       | Stored             |
| Research Project | Keyword          | Simple           | Multi-valued | Non-key            |
| Student          | Student_ID       | Simple           | Single       | <b>Primary Key</b> |
| Student          | Name             | Simple           | Single       | Non-key            |
| Student          | Email            | Simple           | Single       | Non-key            |
| Student          | Date_of_Birth    | Simple           | Single       | Stored             |
| Publication      | Publication_ID   | Simple           | Single       | <b>Primary Key</b> |
| Publication      | Title            | Simple           | Single       | Non-key            |
| Publication      | Publication_Date | Simple           | Single       | Stored             |
| Publication      | DOI              | Simple           | Single       | Non-key            |

| Entity            | Attribute      | Attribute Type | Value  | Key Status  |
|-------------------|----------------|----------------|--------|-------------|
| Publication       | Publisher      | Simple         | Single | Non-key     |
| Project Assistant | Assistant_Name | Simple         | Single | Partial Key |
| Project Assistant | Role           | Simple         | Single | Non-key     |
| Project Assistant | Joining_Date   | Simple         | Single | Non-key     |